The Poisson Random Variable

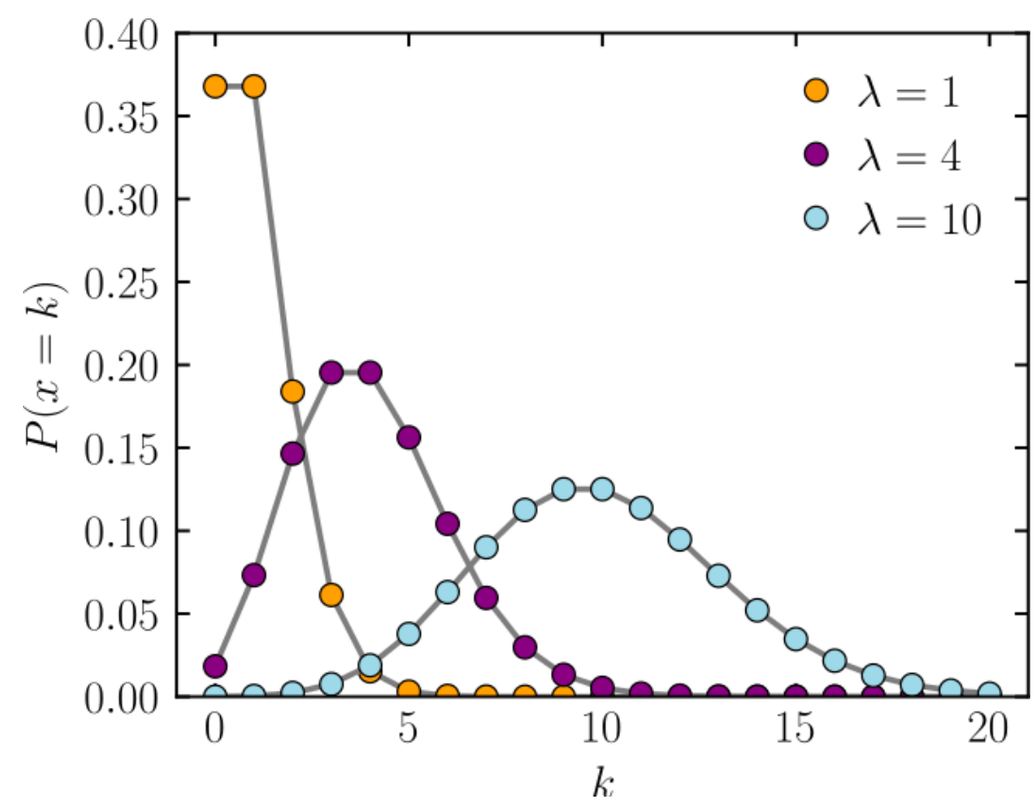
The **Poisson distribution** is the standard model for counting the number of events happening in given interval of time. The key feature is that the events occur independently and with a constant average rate over time.

Examples Number of hospital emergency admissions per night, number of calls at call center per day, number of accidents in a district per month, number of earthquakes in a time window, number of insurance claims filed in a day, number of credit card fraud incidents per day, number of meteor sightings per night,…

$X \in \mathbb{N}_0$

$$p_X(x) = P(X = k) = e^{-\mu} \frac{\mu^k}{k!}$$

μ: central value of the distribution



Poisson Modelling Example: Economic Shocks Over a Decade

Example Suppose significant economic shocks (such as recessions or market crashes) occur at an average rate of 0.3 per year. Over a 10-year period, the total number of shocks is modelled as a Poisson random variable with parameter $\mu = 0.3 \cdot 10 = 3$.

1. What is the probability of observing exactly 3 shocks in 10 years?
2. What is the probability of observing more than 1 shock in 10 years?

X := number of shocks in 10 years $\qquad X \sim \text{Poisson}(3)$

1. $P(X=3) = p_X(3) = e^{-3}\dfrac{3^3}{3!} = \text{dpois}(\overset{k}{3}, \overset{\mu}{3}) = 0.2240$ $\quad P(X=k)$

2. $P(X>1) = \sum_{k=2}^{+\infty} e^{-3}\dfrac{3^k}{k!} = 1 - \underbrace{e^{-3}\dfrac{3^0}{0!}}_{P(X=0)} - \underbrace{e^{-3}\dfrac{3^1}{1!}}_{P(X=1)} = 1 - \text{ppois}(1,3)$

$1 - \boxed{P(X \leq 1)} \rightarrow \text{ppois}(1,3)$

$= 0.80$